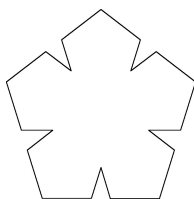




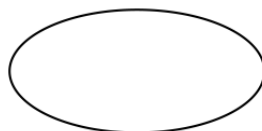
1. Determine symmetries of the following shapes. Write their composition tables.

(a) Square.

(b) Five petaled flower.



(c) Ellipse.



In each of these cases, count those elements in the corresponding symmetry groups which are inverses of themselves.

2. Roboto is one of the fairly symmetric fonts to write Latin. Classify Latin alphabets written in Roboto font according to their symmetry groups.

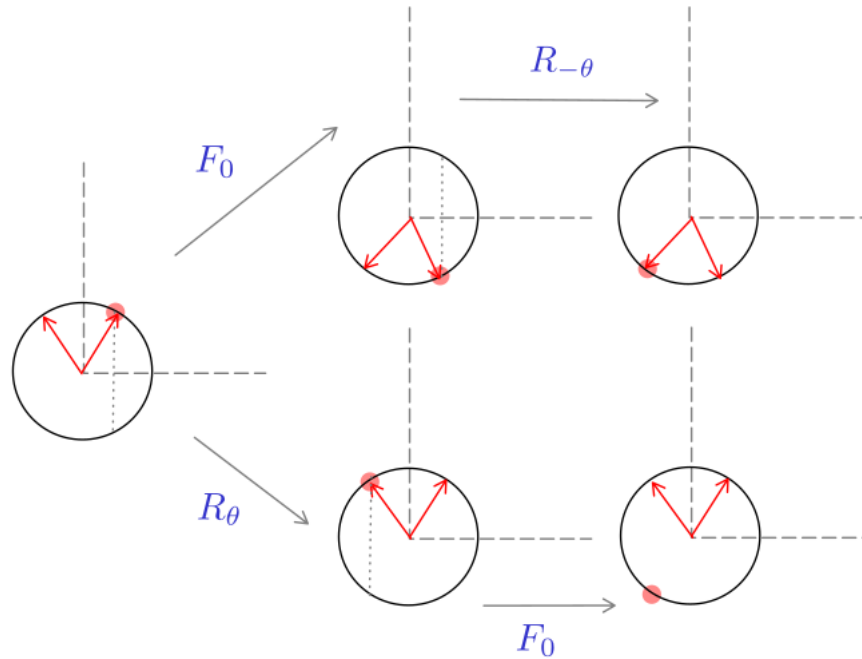
A	B	C	D	E	F	G
H	I	J	K	L	M	N
O	P	Q	R	S	T	U
V	W	X	Y	Z		

3. Let A be a 2×2 orthogonal matrix with entries in $\mathbb{R}$. Show that there is some θ such that either $A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ or $A = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$.
4. Find a 3×3 orthogonal matrix A with entries in $\mathbb{R}$ such that (i) $\det(A) = 1$, (ii). $\det(A) = -1$.
5. Find infinitely many 3×3 orthogonal matrices with entries in $\mathbb{Q}$.

6. Recall the notation.

- (i). T_u : translation by vector $u \in \mathbb{R}^2$.
- (ii). R_θ : rotation by angle θ in anticlockwise direction.
- (iii). F_0 : reflection about x -axis.

The following picture depicts the composition rule $F_0 R_\theta = R_{-\theta} F_0$.



Draw pictures to depict the following composition rules: $T_u T_v = T_{u+v}$, $R_\theta R_\phi = R_{\theta+\phi}$, $F_0^2 = 1$, $R_\theta T_u = T_{u'} R_\theta$ (where $u' = R_\theta(u)$), and $F_0 T_u = T_{u'} F_0$ (where $u' = F_0(u)$).

- 7. *Shoe-socks property.* Let $(X, *)$ be a group and $a, b \in X$. Let a' and b' be inverses of a and b , respectively. Show that $(ab)' = b'a'$.
 - 8. *True or false? Justify.* If $a, b \in \text{sym}(\mathbb{R}^2)$ are self inverses then ab is also a self inverse.
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